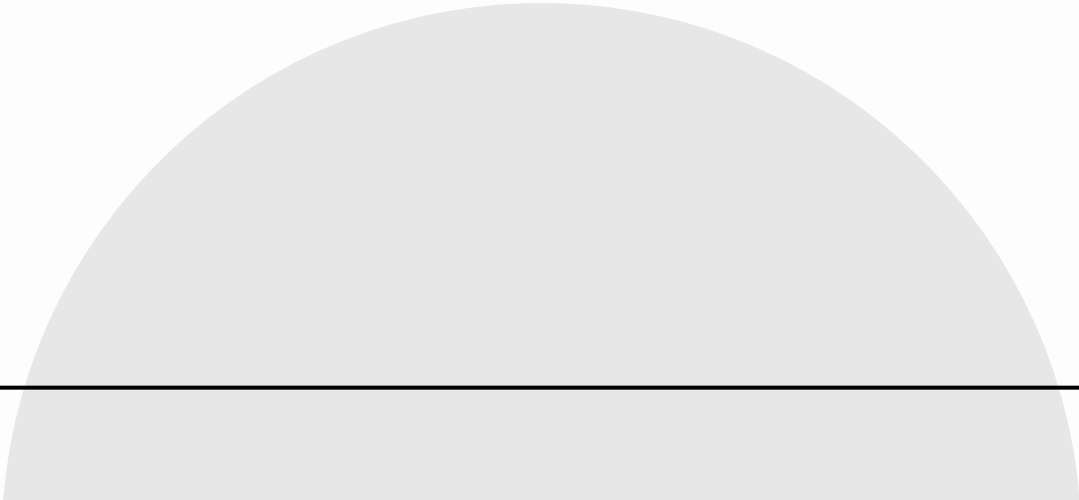


Data Science

Prof. Rayson Laroca
rayson@ppgia.pucpr.br

CORRELATIONS



Related Variables

- Correlation analysis examines the relationship between two or more variables. It helps us identify whether the relationship is:
 - **Direct** or **inverse**
 - **Strong** or **weak**
- **Important:** correlation does **NOT** imply causation

Example

How can we assess
the relationship between
production volume and **total cost**?

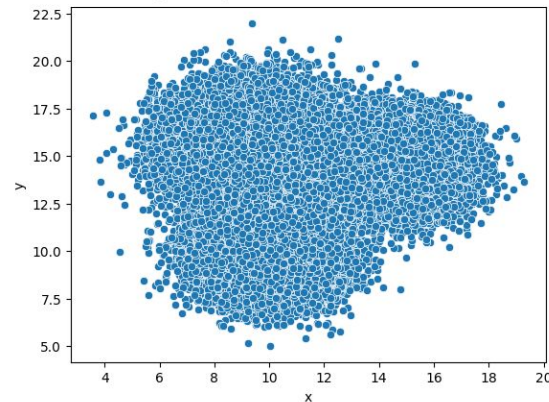
Month	Production (L)	Total Cost (R\$)
Jan	20,200	19,000
Feb	16,700	17,000
Mar	14,800	14,000
Apr	16,000	15,000
May	12,100	14,000
Jun	13,000	15,000
Jul	11,600	13,000
Aug	15,500	16,000
Sep	18,900	18,000
Oct	20,000	19,000
Nov	22,500	20,000
Dec	23,000	21,000

Scatter Plots (Recap)

- Useful for visualizing the relationship between two **numerical variables**
 - Plot individual data points on a **2D plane** (x-axis vs. y-axis)
- Identify patterns and trends, **correlations**, clusters and outliers
- Adjust marker size (and/or transparency) to improve readability

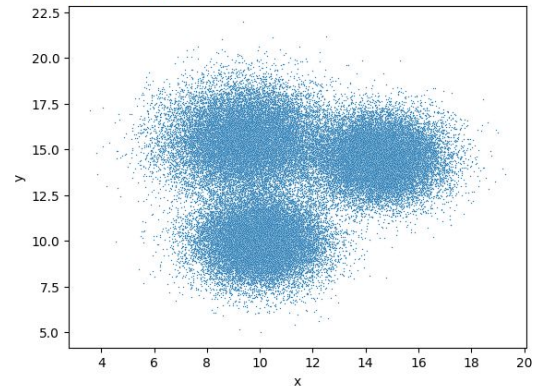
```
sns.scatterplot(data=df, x='x', y='y')
```

<Axes: xlabel='x', ylabel='y'>

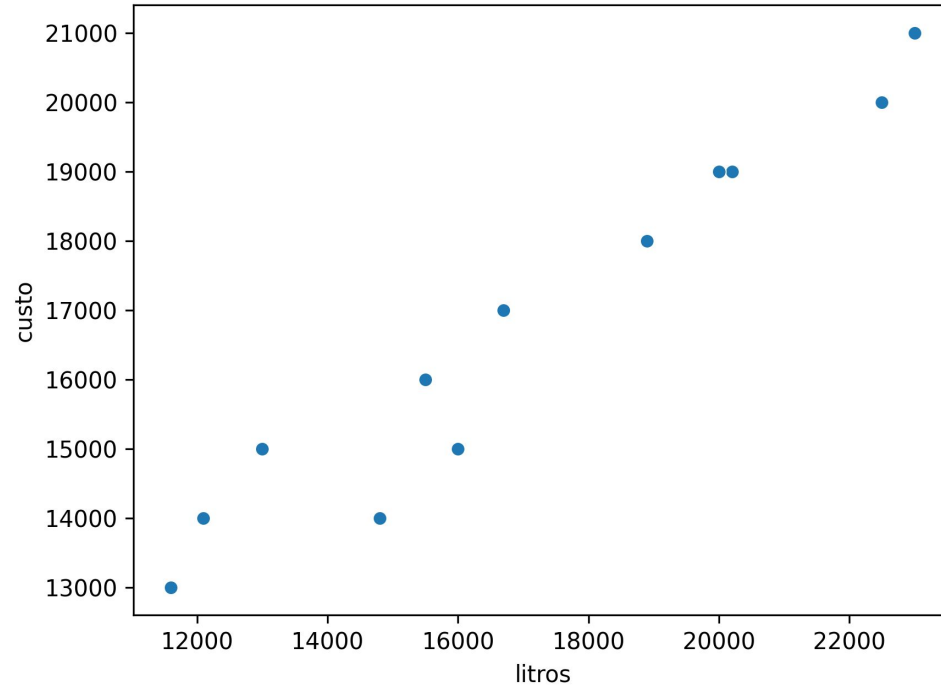


```
sns.scatterplot(data=df, x='x', y='y', s=1)
```

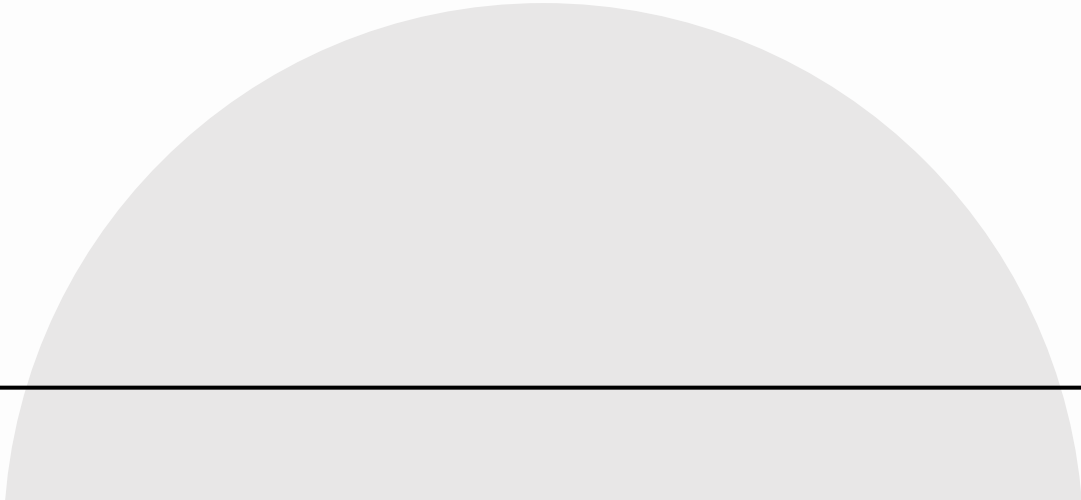
<Axes: xlabel='x', ylabel='y'>



Scatter Plot – Production vs. Total Cost



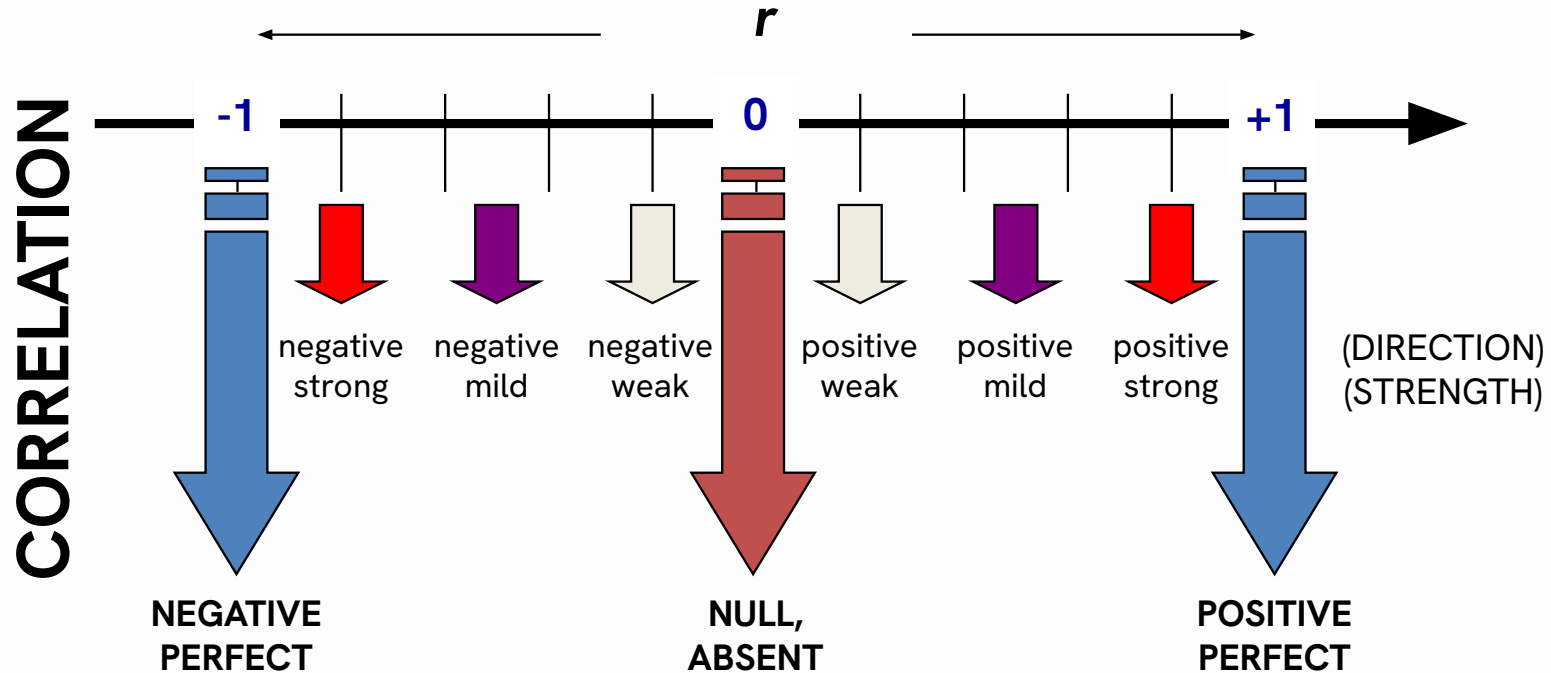
PEARSON'S CORRELATION COEFFICIENT



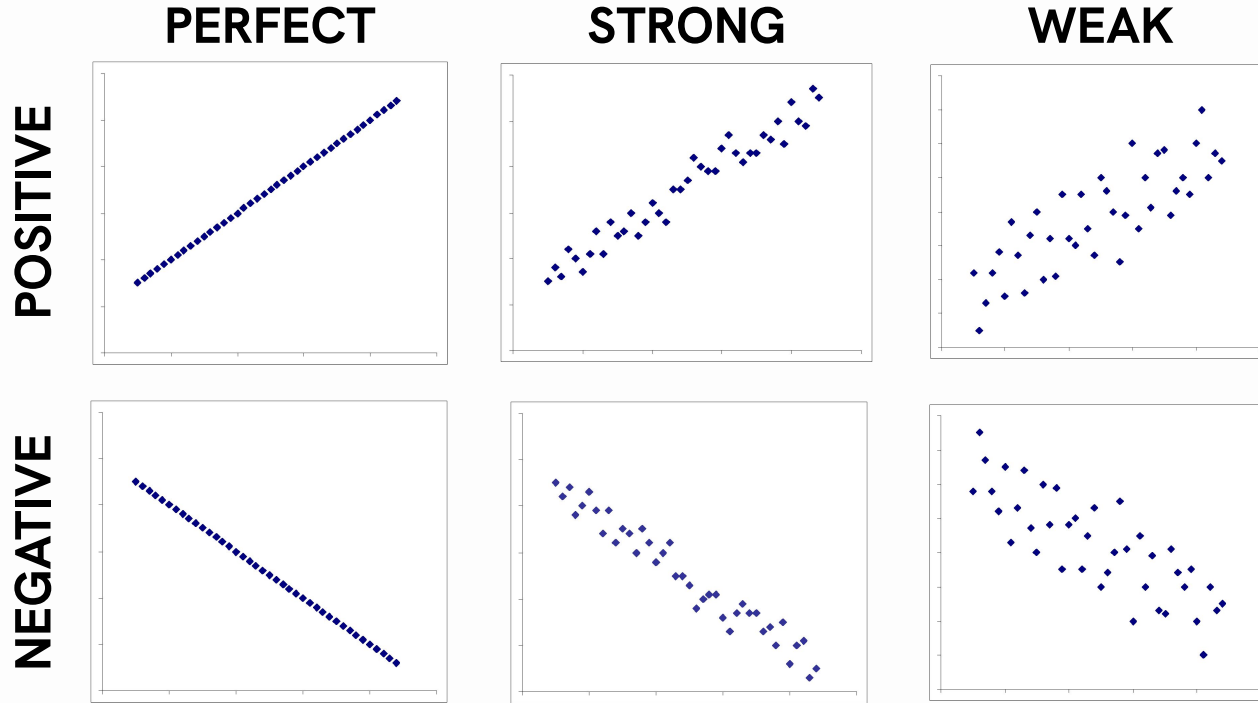
Pearson Correlation Coefficient (r)

- Measures **linear relationships** between two **continuous variables**
- Uses the **original (raw) values** of the variables
 - **Sensitive to outliers** (extreme values can distort results)
- Ranges from **-1** to **+1**
 - **+1**: perfect positive linear relationship
 - **0**: no linear relationship
 - **-1**: perfect negative linear relationship

Interpreting Pearson Correlation (r)

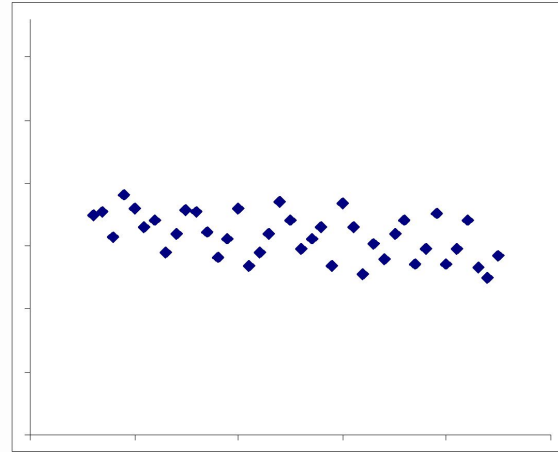
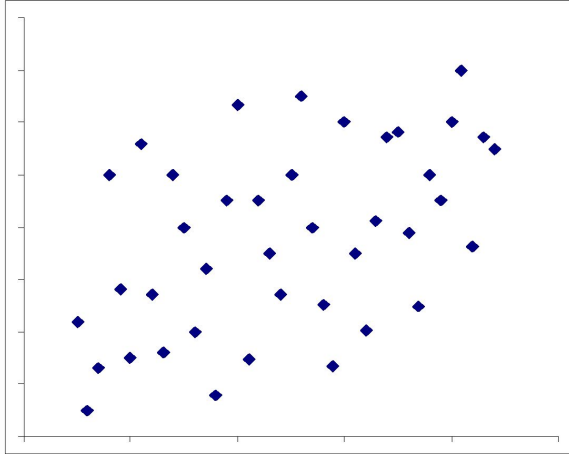


Visual Analysis



Visual Analysis

NULL OR ABSENT CORRELATION



A Practical Rule of Thumb

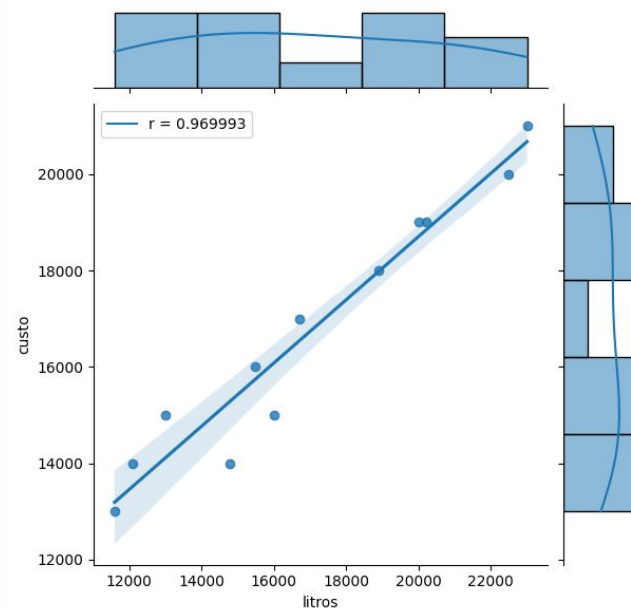
$ r $ (magnitude)	Interpretation
$ r < 0.4$	Weak correlation
$0.4 \leq r < 0.7$	Moderate correlation
$0.7 \leq r $	Strong correlation

Note:

There is no universal agreement on these thresholds.
Different fields may adopt different cutoffs!

Practice

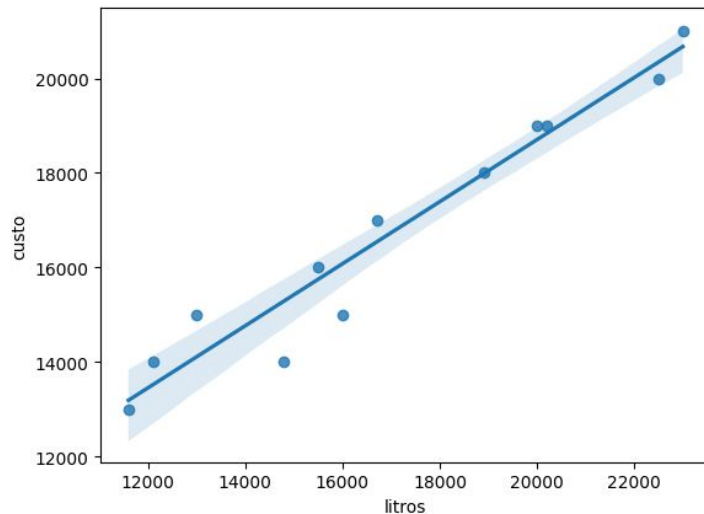
- Let's implement the example of production levels and costs.
 - `df =`
`pd.read_csv('https://raysonlaroca.github.io/teaching/data-science/datasets/meses_producao_custo.csv')`



sns.regplot()

- Useful for visualizing the relationship between two **continuous variables**
- Main features (in this example):
 - **Scatterplot**: displays the individual data points
 - **Regression line**: shows the general trend of the data
 - **Confidence interval**: adds a shaded region around the regression line to represent estimation uncertainty

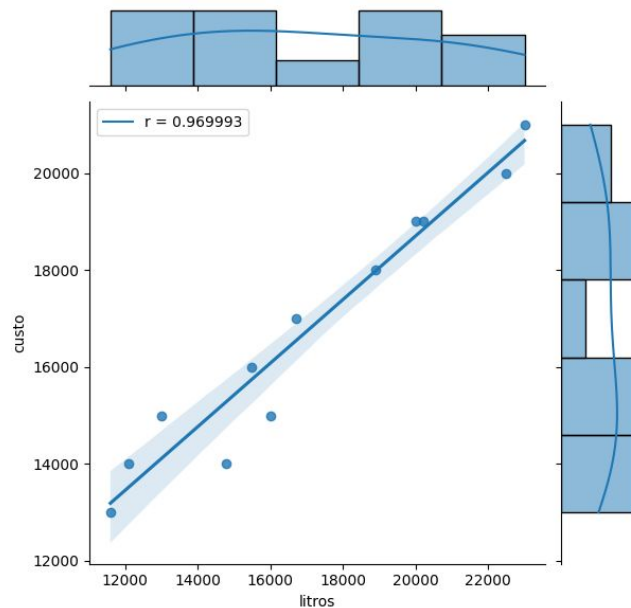
```
sns.regplot(data=df, x='litros', y='custo')
```



sns.jointplot()

- Useful for visualizing the relationship between two **continuous variables**
- Main features (in this example):
 - **Scatterplot**
 - **Regression line**
 - **Confidence interval**
 - **Histograms**: show the distribution of each variable
 - Note: **KDE curves** are overlaid on the histograms

```
p = sns.jointplot(data=df, x='litros', y='custo', kind='reg')  
phantom, = p.ax_joint.plot([], [], linestyle='-', alpha=1)  
p.ax_joint.legend([phantom], [f'r = {r:.6f}'])
```



Computing Pearson Correlation

```
# pandas (pairwise correlation of columns)
df.select_dtypes(include=[np.number]).corr()
```

	litros	custo
litros	1.000000	0.969993
custo	0.969993	1.000000



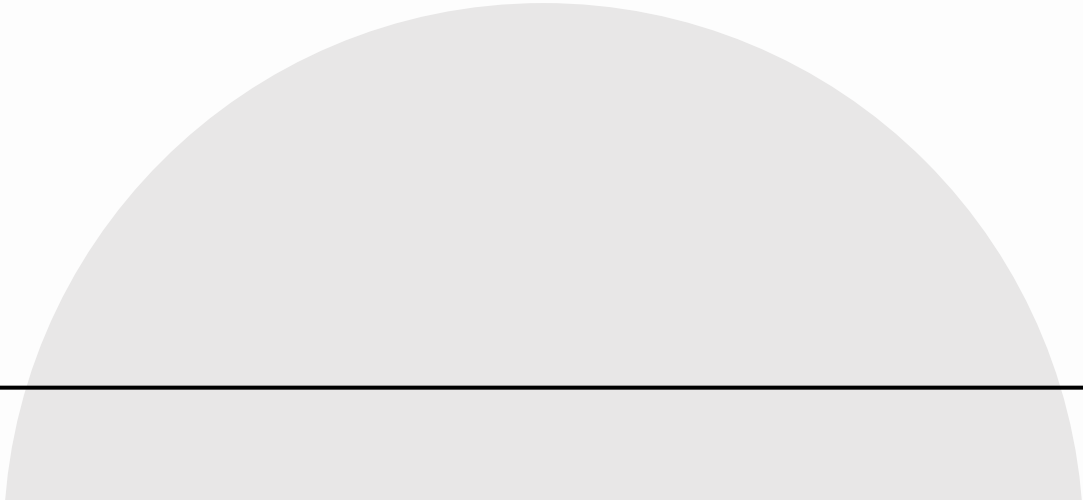
```
# pandas (two columns)
r = df['litros'].corr(df['custo'], method='pearson')
print(r)
```

```
0.9699933926497077
```

```
# scipy (two columns)
r, _ = stats.pearsonr(df['litros'], df['custo'])
print(r)
```

```
0.9699933926497077
```


SPEARMAN'S RANK CORRELATION



Spearman's rank correlation coefficient (ρ)

- Measures the strength and direction of a **monotonic relationship** between two variables
 - As one variable increases, the other tends to **increase or decrease consistently**, though **not necessarily linearly**
- Best used when:
 - the data **does not follow a normal distribution**
 - the relationship is **nonlinear but monotonic**
 - the variables are **ordinal**

Spearman's rank correlation coefficient (ρ)

- Uses **ranked values** rather than original values
 - First, the data for each variable is ranked in ascending or descending order
 - Then, the correlation is calculated using these ranks
- It is **less sensitive to outliers** than Pearson correlation
- Range: -1 to +1 (same general interpretation as Pearson)

Key idea

- Spearman measures whether two variables move in the **same general order**, even when the relationship is **not perfectly linear**:

Original X	Rank of X	Original Y	Rank of Y
10	1	100	1
20	2	200	2
30	3	400	3
40	4	800	4

- If the ranks increase together, the Spearman correlation is **positive**
- If one rank increases while the other decreases, it is **negative**

Computing Spearman Correlation

```
# pandas (pairwise correlation of columns)
df.select_dtypes(include=[np.number]).corr(method='spearman')
```

	litros	custo
litros	1.000000	0.970137
custo	0.970137	1.000000



```
# pandas (two columns)
r = df['litros'].corr(df['custo'], method='spearman')
print(r)
```

```
0.9701365070055414
```

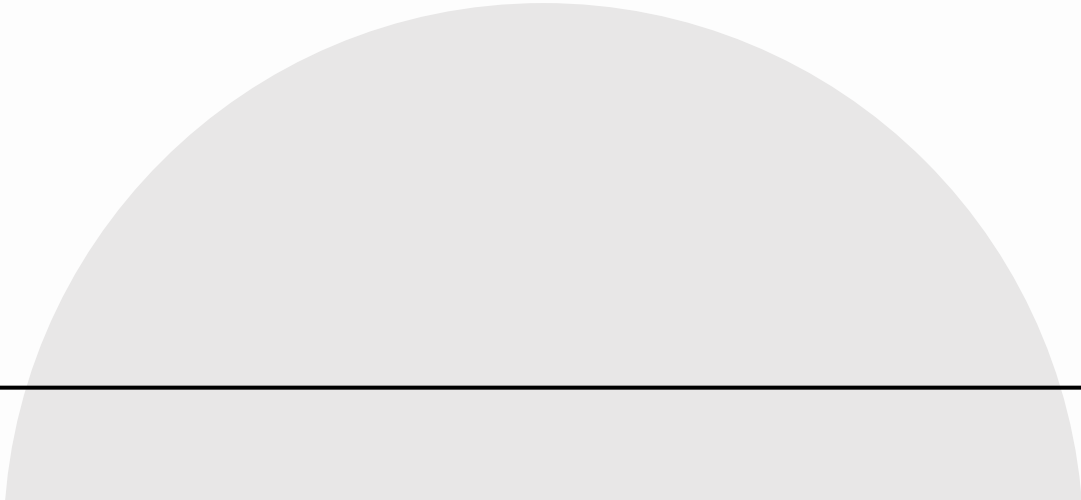
```
# scipy (two columns)
r, _ = stats.spearmanr(df['litros'], df['custo'])
print(r)
```

```
0.9701365070055414
```

Pearson vs. Spearman: Summary

Feature	Pearson	Spearman
Type of relationship	Linear	Monotonic
Based on	Raw values	Ranks
Sensitivity to outliers	Higher	Lower
Typical advantage	Captures linear association well	More robust to outliers and non-normal data

(COR)RELATION BETWEEN CATEGORICAL VARIABLES



Cross-tabulation

- A simple way to examine the relationship between **categorical variables** is to use **cross-tabulations** (also called **contingency tables**)
- **pd.crosstab()** and **heatmaps** help us quickly identify **patterns** and **group differences** in categorical data

Practice

- Let's create a cross-tabulation using the **titanic** dataset
 - `df =`
`pd.read_csv('https://raysonlaroca.github.io/teaching/data-science/datasets/titanic.csv')`

Cross-tabulation

- Summarizes the relationship between **categorical variables**
- Can be used **with two or more columns**
- By default, shows **raw counts**
 - The **normalize** parameter shows **proportions** instead of raw counts

```
ctab = pd.crosstab(df['Sex'], df['Survived'])  
ctab
```

Survived	0	1
Sex		
female	81	233
male	468	109

```
ctab = pd.crosstab(df['Sex'], df['Survived'],  
                   normalize='index')  
ctab
```

Survived	0	1
Sex		
female	0.257962	0.742038
male	0.811092	0.188908

MULTICOLLINEARITY

Multicollinearity

- Occurs when **two or more input variables** are highly correlated with each other
- Why it matters in machine learning models:
 - Makes it hard to interpret the individual effect of each variable
 - Unstable coefficients in linear models
 - Introduces redundant information into the model
 - Tree-based models, such as Random Forests, are usually less affected

Variable Inflation Factor (VIF)

- VIF indicates how much the variance of a coefficient is inflated by correlation with other variables
 - VIF is computed as: $VIF = 1/(1-R^2)$
- **VIF = 1** → no correlation with other variables
- VIF has a **minimum value of 1** and **no upper bound**
- **Higher VIF values** indicate stronger multicollinearity
 - As a rule of thumb, **VIF > 10** often suggests problematic multicollinearity

Handling Multicollinearity

- Typical workflow:
 - Compute the **VIF** for each input variable
 - Identify variables with high **VIF** values (e.g., > 10)
 - Consider **removing**, **combining**, or **transforming** those variables
 - Repeat until all VIF values are below the chosen threshold

Practice

- Let's investigate multicollinearity in practice!
 - `df =`
`pd.read_csv('https://raysonlaroca.github.io/teaching/data-science/datasets/ommlbd_familiar.csv')`

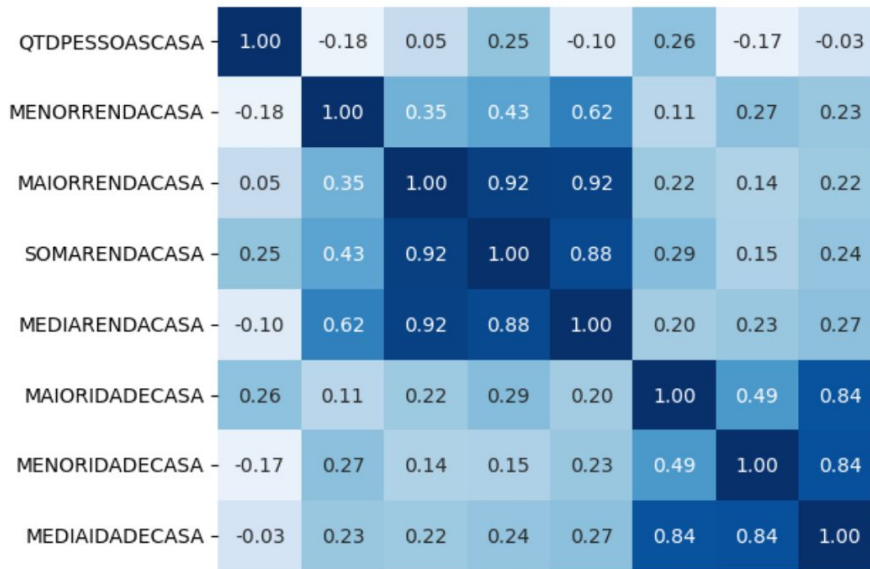
Pairwise Correlation Between Variables

```
corr = df.corr()  
corr
```

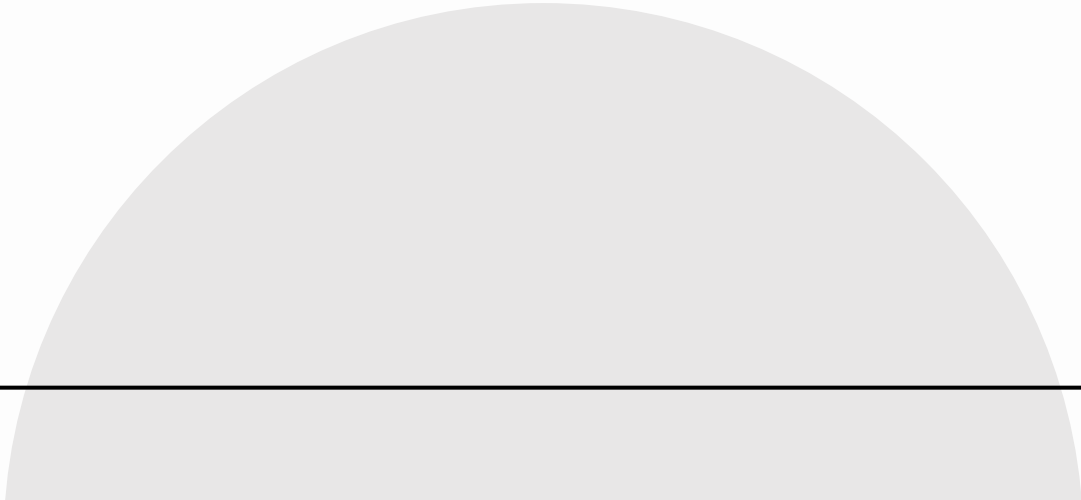
	QTDPESSOASCASA	MENORRENDACASA	MAIORRENDACASA	SOMARENDACASA	MEDIARENDACASA
QTDPESSOASCASA	1.000000	-0.183627	0.053948	0.252608	-0.103643
MENORRENDACASA	-0.183627	1.000000	0.352656	0.433097	0.615976
MAIORRENDACASA	0.053948	0.352656	1.000000	0.923269	0.917403
SOMARENDACASA	0.252608	0.433097	0.923269	1.000000	0.880107
MEDIARENDACASA	-0.103643	0.615976	0.917403	0.880107	1.000000
MAIORIDADECASA	0.259603	0.106155	0.219026	0.285825	0.198684
MENORIDADECASA	-0.166085	0.269575	0.144396	0.145600	0.230392
MEDIAIDADECASA	-0.033299	0.232897	0.218736	0.244595	0.269323

Pairwise Correlation Between Variables (Heatmap)

```
sns.heatmap(corr, annot=True, fmt='.2f', cmap='Blues')  
plt.show()
```



CORRELATION AND CAUSATION



Correlation Does Not Imply Causation

- We often observe a correlation between two variables and assume a cause-and-effect relationship
- This may be true, but it is **not necessarily the case**
 - Correlation can also be explained by:
 - Coincidence
 - A third variable affecting both
 - Many other factors
- Always interpret correlations with caution!

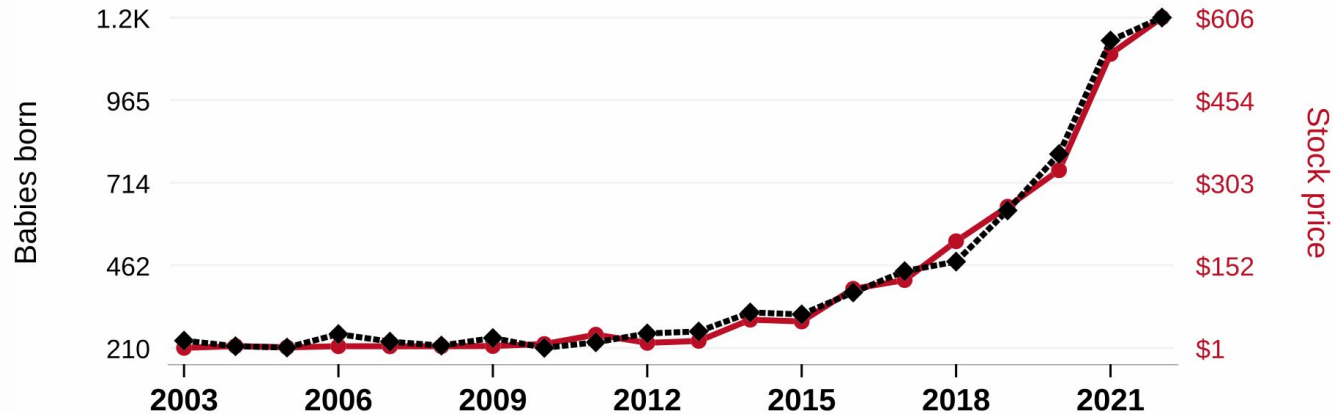
Correlation Does Not Imply Causation

- An observed correlation between two variables **does not mean** that one causes the other.
- **Example:** ice cream sales increase as the temperature increases during summer, and so does the sales of fans
 - These two variables may be **positively correlated**
 - However, this does **not** mean that buying ice cream causes people to buy fans
- **Takeaway:**
 - A third variable, such as **temperature**, may influence both

Popularity of the first name Stevie

correlates with

Netflix's stock price (NFLX)



◆--- Babies of all sexes born in the US named Stevie · Source: US Social Security Administration

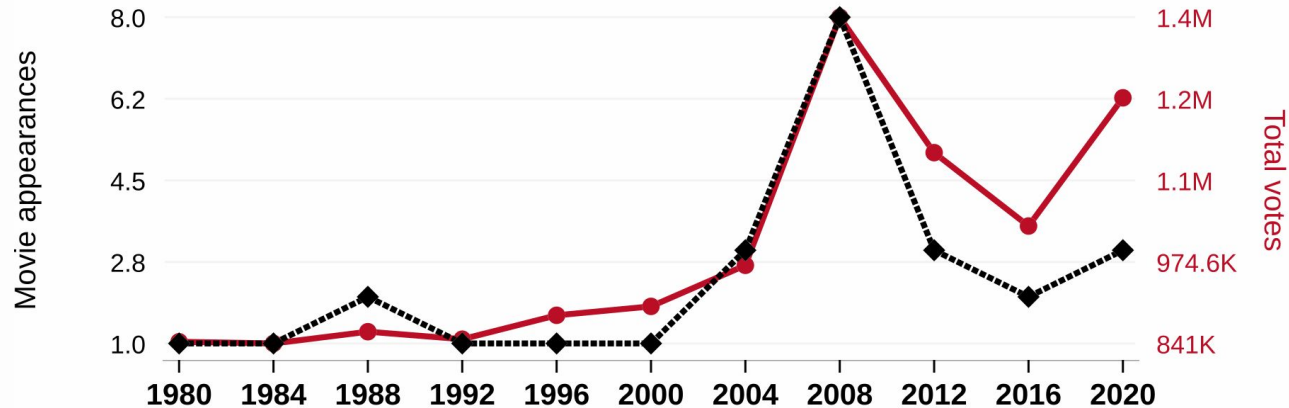
●— Opening price of Netflix (NFLX) on the first trading day of the year · Source: LSEG Analytics (Refinitiv)

2003-2022, $r=0.996$, $r^2=0.993$, $p<0.01$ · tylervigen.com/spurious/correlation/3268

The number of movies Robert Downey Jr. appeared in

correlates with

Votes for the Democratic Presidential candidate in Indiana



◆ The number of movies Robert Downey Jr. appeared in · Source: The Movie DB

● Total votes cast for the Democrat Presidential candidate in Indiana · Source: MIT Election Data and Science Lab, Harvard Dataverse

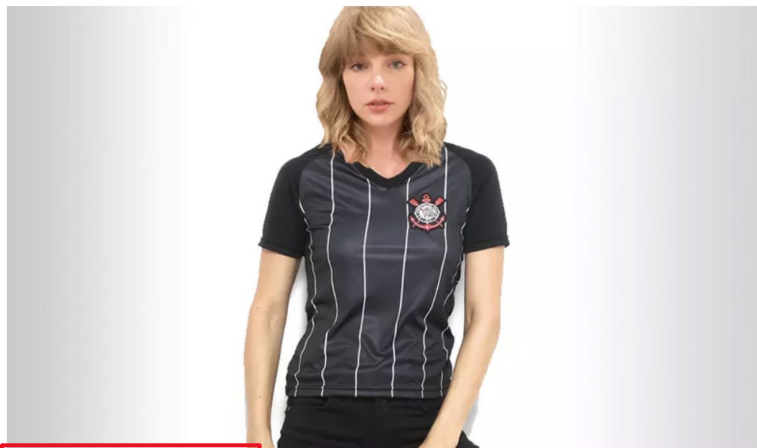
1976-2020, $r=0.871$, $r^2=0.758$, $p<0.01$ · tylervigen.com/spurious/correlation/5861

Taylor Swift é corintiana? Time nunca perdeu quando cantora lança álbum

Anna Satie, da CNN em São Paulo

11 de dezembro de 2020 às 15:33 | Atualizado 11 de dezembro de 2020 às 15:43

Ouvir



Será que Taylor Swift é corintiana?

Foto: CNN Brasil/Reprodução/Instagram

O **lançamento surpresa** de Evermore, nono álbum de Taylor Swift, nesta sexta-feira (11) foi motivo de alegria para os torcedores do Corinthians.

O porquê? O time nunca perdeu uma partida imediatamente antes ou depois da cantora apresentar um novo CD. Por um terço das vezes, o alvinegro paulista empatou, mas a maior parte das partidas terminou em vitória corintiana.

Compartilhar



MAIS DA CNN BRASIL



Após embate com Pazuello, Doria escala técnicos para falar...

Ouça os episódios do 'O que eu faço?'



Quem já teve Covid-19 precisará tomar vacina?

Vacina da Covid-19 é contraindicada para menores de 18 e...

MAIS DA CNN BRASIL



Planalto pressionou Pazuello a mudar estratégia da vacina em reunião na segunda



Evidence That Facebook Cancelled Out the Cholesterol-Lowering Effects of Justin Bieber

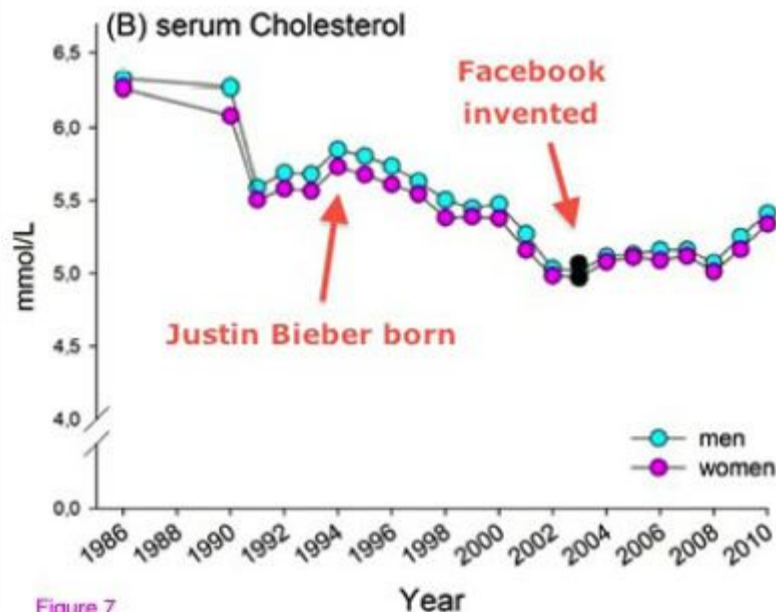
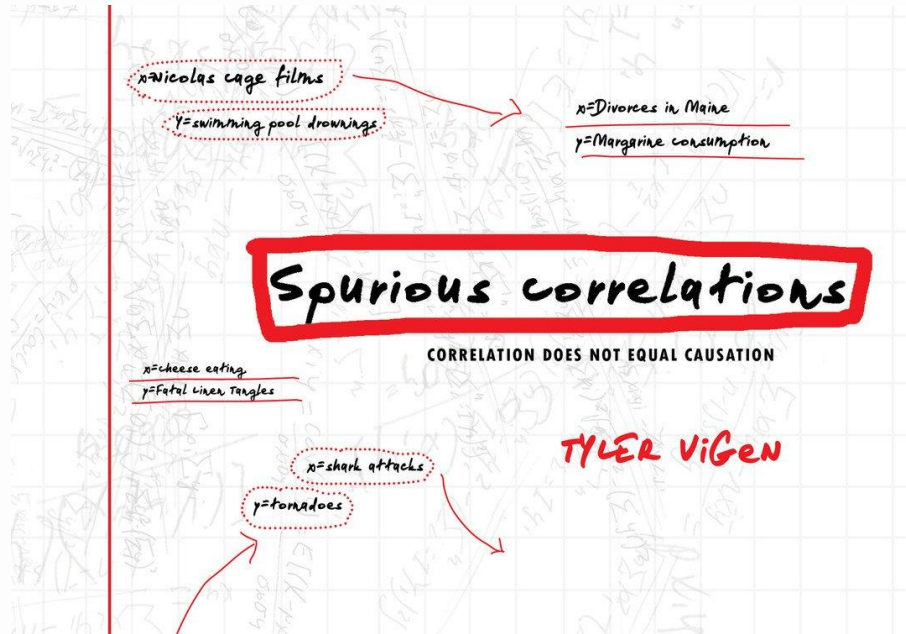


Figure 7

More examples: Spurious Correlations

- <https://www.tylervigen.com/spurious-correlations>



Project

- Use the remainder of today's class to work on your project.